

Redline lab Walkthrough

As a member of the Security Blue team, your assignment is to analyze a memory dump using Redline and Volatility tools. Your goal is to trace the steps taken by the attacker on the compromised machine and determine how they managed to bypass the Network Intrusion Detection System (NIDS). Your investigation will identify the specific malware family employed in the attack and its characteristics. Additionally, your task is to identify and mitigate any traces or footprints left by the attacker.

Q1.What is the name of the suspicious process?

To see suspicious processes we use pstree to analyze all the processes in the memory file.

```
PS C:\Users\analyst > vol -f C:\Users\analyst\Desktop\temp_extract_dir\MemoryDump.mem windows.pstree | Out-file C:\Users\analyst\Desktop\temp_extract_dir\pstree.txt
```

```
\\Device\\HarddiskVolume3\\Users\\Tammam\\AppData\\Local\\Temp\\c3912af058\\oneetx.exe
```

Looking into the pstree output, find file path that is quite odd such as the picture above on user tammam temp folder contains a strange .exe file.

Q2.What is the child process name of the suspicious process?

PID	PPID	Process Name	Architecture
5896	8844	oneetx.exe	0xad81
* 7732	5896	rundll32.exe	0xad81

This is the child process because the Parent PID of that file is the same as the PID by the oneetx.exe which means the rundll32.exe is created by oneetx.exe file. Refer to PID 5896

rundll32.exe

Q3.What is the memory protection applied to the suspicious process memory region?

PID	Process	Offset	Start VPN	End VPN	Tag	Protection	CommitCharge	PrivateMemory	Parent	File	File output
5896	oneetx.exe	0xfffffad818ddb2c40	0x73da0000	0x74024fff	Vad	PAGE_EXECUTE_WRITECOPY	554	0	0x0	\Windows\SysWOW64\AcLayers.dll	Disabled
5896	oneetx.exe	0xfffffad818ddc5200	0x1a60000	0x2e60fff	Vad	PAGE_READONLY	0	0	0	0xfffffad818ddb2c40	N/A
5896	oneetx.exe	0xfffffad818ddc5b70	0x1390000	0x139ffff	VadS	PAGE_READWRITE	10	1	0	0xfffffad818ddc5200	N/A
5896	oneetx.exe	0xfffffad818ddc6b20	0x1000000	0x11fffff	VadS	PAGE_READWRITE	11	1	0	0xfffffad818ddc5b70	N/A
5896	oneetx.exe	0xfffffad818ddab1c0	0xfc0000	0xfc0fff	Vad	PAGE_READONLY	0	0	0	0xfffffad818ddc6b20	N/A
5896	oneetx.exe	0xfffffad818d2c7ac0	0xec0000	0xf7fff	Vad	PAGE_EXECUTE_WRITECOPY	0	0	0	0xfffffad818ddab1c0	\Users\Tammam\AppData\Local\
5896	oneetx.exe	0xfffffad818d65940	0x400000	0x437fff	VadS	PAGE_EXECUTE_READWRITE	56	1	0	0xfffffad818d2c7ac0	N/A
5896	oneetx.exe	0xfffffad818ddad20	0xfe0000	0xfe0fff	Vad	PAGE_READONLY	0	0	0	0xfffffad818d65940	N/A

PID	Process	Start VPN	End VPN	Tag	Protection	CommitCharge	PrivateMemory	File output	Notes	Hexdump	Disasm
5896	oneetx.exe	0x400000	0x437fff	VadS	PAGE_EXECUTE_READWRITE	56	1	Disabled	MZ header		

Both output malfind and vadinfo show this, This output shows that **oneetx.exe (PID 5896)** has a suspicious private memory region containing a Windows executable image. This is commonly called **RWX memory**. It is suspicious because malware can write payload code into the region and execute it directly.

PAGE_EXECUTE_READWRITE

Q4.What is the name of the process responsible for the VPN connection?

State	Local Address	Foreign Address	Process
CLOSED	5896	oneetx.exe	
CLOSED	4628	tun2socks.exe	

netstat shows **tun2socks.exe PID 4628** directly making the network connection.

PID	PPID	Process	PrivateMemory
6724	3580	Outline.exe	0xad818
4224	6724	Outline.exe	0xad818
4628	6724	tun2socks.exe	0xad818

psutil shows that **PID 4628 was launched by Outline.exe PID 6724**.

Therefore, **Outline.exe is the main VPN application**, while tun2socks.exe is its helper process that handles the tunnel traffic.

Q5.What is the attacker's IP address?

Protocol	Local Address	Local Port	Foreign Address	Foreign Port	Process	Time
DPv6	::	5353	*	0	5328 msedge.exe	2023-05-21 23:
CPv4	10.0.85.2	55462	77.91.124.20	80	CLOSED 5896 oneetx.exe	

Use netstat command and find the suspicious exe file process, then refer to its ip.

77.91.124.20

Q6.What is the full URL of the PHP file that the attacker visited?

Use strings matching the attacker ip to find what url exists in the memory.

```
PS C:\Users\analyst > strings64.exe -n 6 `
>> C:\Users\analyst\Desktop\temp_extract_dir\MemoryDump.mem |
>> Select-String -SimpleMatch "77.91.124.20"

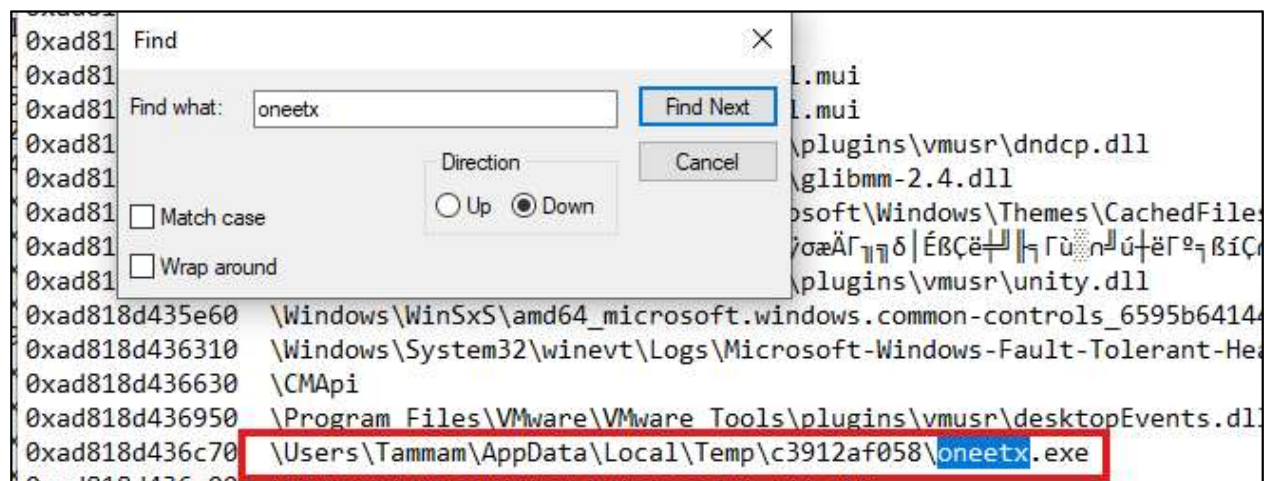
http://77.91.124.20/ E
77.91.124.20/stor
http://77.91.124.20/store/games1
http://77.91.124.20/store/games/i
http://77.91.124.20/store/games/Plugins/clip64.dll
http://77.91.124.20/DSC01491/foto0195.exe
77.91.124.20
http://77.91.124.20/ E
http://77.91.124.20/DSC01491/
77.91.124.20
http://77.91.124.20/store/games/index.php
http://77.91.124.20/DSC01491/
```

http://77.91.124.20/store/games/index.php

Q7.What is the full path of the malicious executable?

```
vol -f .\MemoryDump.mem windows.filescan | Out-File
C:\Users\analyst\Desktop\temp_extract_dir\filescan.txt
```

Use this command to export every file and its path, but basically the **pstree** output already show the file path, **filescan** is worth checking if you need to **find or recover an actual file** rather than merely identify a process.



C:\Users\Tammam\AppData\Local\Temp\c3912af058\oneetx.exe